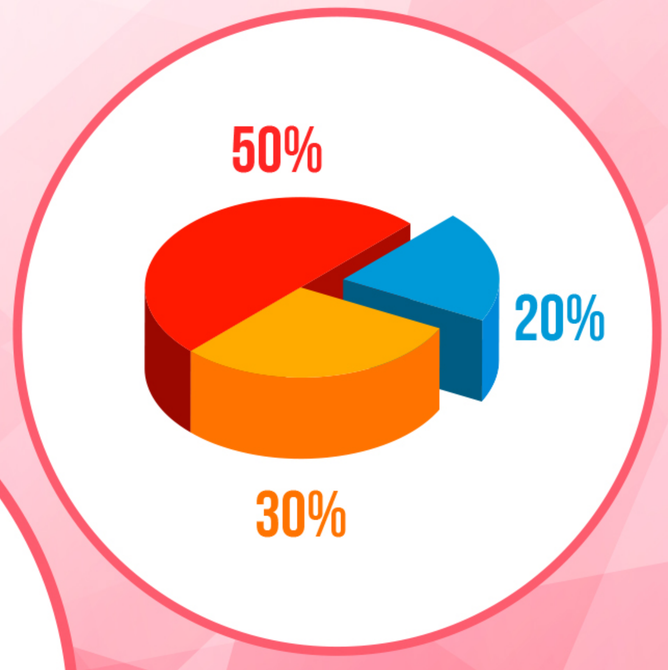
DATA

Tables & other important Concepts for DI

Assignment 04



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DI: Tables – 04

DIRECTIONS ***for questions 1 – 4:*** *Refer to table given below and answer the questions that follow.*

Five machines produce fuses that are used in four wheeled vehicles. The fuses are inspected and either accepted or rejected. The following table gives the number of fuses accepted and the percentage of the total production rejected for each machine.

Machine no.	*Number Accepted*	*Percentage of production rejected*
1	400	20
2	380	22
3	380	16
4	400	18
5	430	12

1. What is the production of machine – 1?

1. 480 2. 500 3.600 4. None of these

2. What is the approximate number of fuses rejected from the production of machine 5?

1. 59 2. 36 3.52 4. 48

3. The total production is highest for machine number

1. 1 2. 2 3.3 4. 5

4. The total production is lowest for machine number

1. 1 2. 2 3.3 4. 4 or 5

DIRECTIONS ***for questions 5 – 7:*** *Refer to the following table to answer the questions that follow.*

	%of Protein	% of Carbohydrates	% of Fat	Cost per 100 Grams
Food A	10	20	30	Rs. 1.80
Food B	20	15	10	Rs. 3.00
Food C	20	10	40	Rs. 2.75

5. If you purchase x grams of food A, y grams of food B, and z grams of food C, the cost will be

1. $\left(\frac{9}{5}x + 3y + \frac{11}{4}z\right)$ ps
2. Rs. (1.8x + 3z +2.75y)
3. (3x + 1.8y + 2.75z) ps
4. Rs. (x + y + z)

6. Which of the following diets would supply the most grams of protein?

1. 500 grams of A
2. 250 grams of B
3. 200 grams of B and 200 grams of C
4. 350 grams of C

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7. All of the following diets would supply at least 75 grams of fat. Which of the diets costs the least?

1. 200 grams of A + 150 grams of B
2. 300 grams of A
3. 200 grams of C
4. 150 grams of A + 100 grams of C

DIRECTIONS for *questions 8 – 10:* *Read the following table and answer the questions that follow.*

Railway Timetable			
City	**Arrival**	**Departure**	**Cumulative Mileage**
A	----------	9:00 am	0
B	9:15 am	9:20 am	12
C	10:40 am	10:55 am	100
D	11:25 am	11:30 am	140
E	12:15pm.	12:25pm	185
F	12:40pm	12:45 pm	190
G	1:45 pm	2:00 pm	250
H	2:30 pm	----------	275

8. What was the average overall speed for the trip, stopping time included?

1. 48 mph
2. 45 mph
3. 50 mph
4. 56mph

9. What was the greatest distance between two consecutive stops?

1. 45 miles
2. 88 miles
3. 100 miles
4. 275 miles.

10. What was the average distance travelled between stops, approximately?

1. 25 miles
2. 42 miles
3. 45 miles
4. 39 miles

Answer Key & Explanations

Sr. No.	Key	Explanation
1	2	In this set, quickly calculate the total production of fuses from the given data; for eg. In case of machine 1 no. accepted is 400 & %age of rejection is 20% which means 400 is 80% of total capacity. Then the total capacity can be calculated as 80/100 of Total Capacity = 400 Machine 1's Total Capacity = 4000/8 = 500 fuses. Machine 2's Total Capacity = 487 fuses Machine 3's Total Capacity = 452 fuses Machine 4's Total Capacity = 487 fuses Machine 5's Total Capacity = 488 fuses By Visual Observation, the production of machine – 1 is 500.
2	1	Number of fuses rejected from the production of machine 5=12% of 488= 58.6 = 59% (approx.)
3	1	In this set, quickly calculate the total production of fuses from the given data; for eg. In case of machine 1 no. accepted is 400 & %age of rejection is 20% which means 400 is 80% of total capacity. Then the total capacity can be calculated as 80/100 of Total Capacity = 400 Machine 1's Total Capacity = 4000/8 = 500 fuses. Machine 2's Total Capacity = 487 fuses Machine 3's Total Capacity = 452 fuses Machine 4's Total Capacity = 487 fuses Machine 5's Total Capacity = 488 fuses By Visual Observation, The total production is highest for machine number 1.
4	3	In this set, quickly calculate the total production of fuses from the given data; for eg. In case of machine 1 no. accepted is 400 & %age of rejection is 20% which means 400 is 80% of total capacity. Then the total capacity can be calculated as 80/100 of Total Capacity = 400 Machine 1's Total Capacity = 4000/8 = 500 fuses. Machine 2's Total Capacity = 487 fuses Machine 3's Total Capacity = 452 fuses Machine 4's Total Capacity = 487 fuses Machine 5's Total Capacity = 488 fuses By Visual Observation, The total production is lowest for machine number 3.
5	1	100g of Food A costs = Rs. 1.80 X gm of Food A costs = 1.80x/100 = 1.08 paise => 9x/5 paise. Similarly; Y gm of Food B costs = 3y paise Z gm of Food C costs = 1¼ z paise. Hence total cost = 9x/5+ 3y + 11z/4.
6	3	500 gm of A supplies $\frac{10}{100}\times 500 = 50\ gm$ of protein 250 gm of B supplies $\frac{20}{100}\times 250 = 50\ gm$ of protein 200 gm of B supplies 40g & 200 gm of C supplies 40g of C $\therefore$ a total of 80 gm of protein. 350 gm of C supplies 70 gm of proteins. Hence option 3.
7	2	i) 200 gm of A costs = 1.8 × 2 = Rs. 3.6 ; 150 gm of B costs = Rs. 4.5 ; $\therefore$Total cost = Rs. 8.10 ii) 300 gm of A costs = Rs. 1.8 × 3 = Rs. 5.40; iii) 200 gm of C costs = Rs. 2.75 × 2 = Rs. 5.50 iv) 150 gm of A costs = Rs. 2.70; 100 gm of C costs = Rs. 2.75; $\therefore$Total cost = Rs. 5.45

		Least cost is that of 300 gm of A.
8	3	Train started at 9.00 a.m. reached the final station at 2.30 p.m., i.e. train traveled for 5 1/2 hrs, including stopping time. During the journey the distance covered is 275 mile. Hence the speed of the train, stopping time included is 275/ 5 1/2 = (275 × 2)/11 = 550/11 = 50 mph. ∴Answer is (3).

9	2	Distance between A and B is 12 – 0 = 12; between B and C is 100 - 12 = 88; between C and D is 140 - 100 = 40,
		between D and E is 185 - 140 = 45;
		between E and F is 190 - 185 = 5;
		between F and G is 250 - 190 = 60;
		and that between G and H is 275 - 250 = 25.
		The route is A- B - C - D -E - F - G - H.
		Hence greatest distance between two consecutive stops is 88 miles. ∴Answer is (2).

10	4	Total distance traveled is 275 miles, while the number of stops from A to H are 7. Hence the average distance per stop is 275/7 = 39.28 = 39 miles. ∴Answer is (4).